

Available triggers (standard, unless otherwise noted)

Trigger type	Channels available on	Description
Edge	Channels 1-8, digital, line, aux	Triggers on a specified slope (rising, falling or alternating between rising and falling) and voltage level on any channel or auxiliary trigger.
Edge transition	Channels 1-4	Triggers on rising or falling edges that cross two voltage levels in > or < the amount of time specified. Edge transition setting from 75 ps to 10 s
Edge then edge (time)	Channels 1-4, digital	The trigger is qualified by an edge. After a specified time-delay between 1.5 ns to 20 s, a rising or falling edge on any one selected input will generate the trigger
Edge then edge (event)	Channels 1-4, digital	The trigger is qualified by an edge. After a specified delay between 1 to 65,000,000,000 rising or falling edges, another rising or falling edge on any one selected input will generate the trigger
Pulse width	Channels 1-4, digital	Triggers on a pulse that is wider or narrower than the other pulses in your waveform by specifying a pulse width and a polarity. Pulse width range settings 75 ps to 20 s. Trigger point can be configured for "end of pulse" or "time out"
Glitch	Channels 1-8, digital	Triggers on glitches narrower than the other pulses in your waveform by specifying a width less than your narrowest pulse and a polarity. Glitch range settings: < 75 ps to < 10 s
Runt	Channels 1-4	Triggers on a pulse that crosses one threshold but fails to cross a second threshold before crossing the first again. Can be time qualified with a range of 75 ps to 10 s
Timeout	Channels 1-4, digital	Triggers the oscilloscope when the waveform has been at a higher voltage than the voltage specified by the Level control for too long (High Too Long), when the waveform has been at a lower voltage than the Level voltage for too long (Low Too Long), or when the waveform has taken too long to pass through the Level voltage (Unchanged Too Long). Timeout settings from 75 ps to 20 s.
Pattern/State	Channels 1-4, digital	Identifies a trigger condition by looking for a specified pattern or a pattern and an edge (state) across the input channels
Setup / hold	Channels 1-4	Triggers on violations of setup time, hold time, or both setup and hold time. Setup times from 75 ps to 20 s and hold times from 75 ps to 100 ns.
Window	Channels 1-4	Specifies a voltage range and then trigger when the waveform either exits this range, enters this range, stays outside the range for too long or too short, or stays inside the range for too long or too short. Range setting from 75 ps to 20 s.
Protocol	Bus dependent	Trigger on certain packets or patterns in protocol-based data. Requires a protocol trigger/decode option, for example D9010LSSP
Generic Protocol	Channels 1-8	Software triggers on NRZ or 8b/10b-encoded data up to 6 Gbps, up to 80-bit pattern. Support multiple clock data recovery methods including constant frequency, 1st-order PLL, 2nd-order PLL, explicit clock, explicit 1st-order PLL, explicit 2nd-order PLL, Fibre Channel, FlexRay receiver, FlexRay transmitter
Burst	Channels 1-4	Triggers on the Nth edge of a burst that occurs after an idle time from 1.5 ns to 20 s.
Nth Edge	Channels 1-8	Triggers on the Nth edge
OR'd Edges	Channels 1-4	Identifies a trigger condition by looking for selected edges on up to four channels
InfiniiScan Zone	Channels 1-8	Qualified trigger across up to 8 user-drawn zones. For each zone, user specifies "must intersect" or "must not intersect." Zones can be drawn on analog channels and combined using Boolean logic. <i>Requires option D9010SCNA</i>
Measurement limit	Channels 1-8, digital, line, aux	Software triggers on the results of the measurement values. For example, when the "time interval error (TIE)" is measured, InfiniiScan can trigger on a specific TIE value. <i>Requires option D9010SCNA</i>
Non-monotonic edge	Channels 1-8	Software triggers on the non-monotonic edge. The non-monotonic edge is specified by setting a hysteresis value. <i>Requires option D9010SCNA</i>